CivicMitra (CivicOrbit)

Assignment 2 - System Design and Initial Prototype

Deliverables 1 to 5

1. System Architecture Diagram

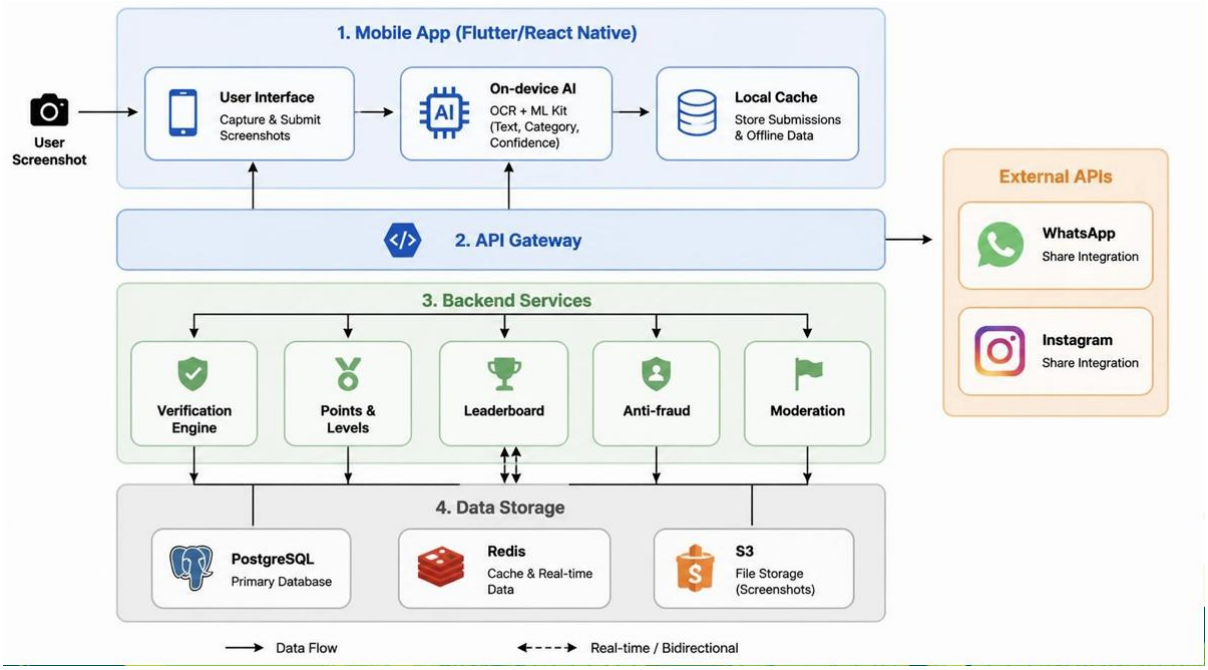


Figure 1: System architecture of CivicMitra

2. Module List & Description Sheet

Module Name	Purpose	Inputs	Outputs	Owner
Authentication	Register and log in users	Email, username, password	Authenticated session, user profile	[Team member]
Dashboard	Show user progress at a glance	User profile, points, completions	Streak, points, impact cards, daily challenges, rank progress	[Team member]
Challenges	List and filter eco challenges	Search text, category filter	Challenge cards with category, points, difficulty	[Team member]
Proof Verification (AI)	Verify proof photo for a challenge	Proof image, challenge title, proof instructions	Verified flag, score, reason	[Team member]
AI Screen Scanner	Match a captured screen/photo to an active challenge	Screen grab, camera photo or uploaded image	Matched challenge, confidence, verified flag, reason	[Team member]
Points, Levels & Badges	Reward verified actions	Verified completions, total points	Points, level, badges, impact values	[Team member]
Leaderboard	Rank users by points	Users' points	Top 50 ranked users	[Team member]
Community Feed	Share eco journey with the community	Post text, photo, likes, comments	Global / Following feed with verified posts	[Team member]
Events	Show and manage eco events	Event details, registrations	Event list ordered by start date	[Team member]

Module Name	Purpose	Inputs	Outputs	Owner
Profile & Settings	Manage user details and preferences	Profile details, settings toggles	Updated profile and settings	[Team member]
Daily Quiz	Test eco knowledge	Quiz answers	Quiz score, quiz badges	[Team member]
Admin	Manage platform content	Challenges, quiz questions	Updated challenges and quizzes	[Team member]

3. Technology Stack Justification

Frontend

- React 19 with TypeScript - reusable components and type safety
- Vite - fast build and development server
- Tailwind CSS - quick, consistent styling
- React Router - page navigation with lazy loading

Backend

- Node.js with Express - local development server and API routes
- Serverless API functions on Vercel (verify-proof, screen-scan, health) - Gemini API key stays on the server

Database

- Firebase Firestore - real-time data for feed, leaderboard and completions
- Firebase Authentication - managed and secure login
- Firestore security rules - control access to data

AI

- Google Gemini (@google/genai) - image-based proof verification and screen scanning, returns structured JSON

Tools

- GitHub - version control
- Vercel - deployment
- Google AI Studio - app development

4. Initial Prototype

- **GitHub repository link:** https://github.com/Project-Based-Learning-Project/CivicMitra

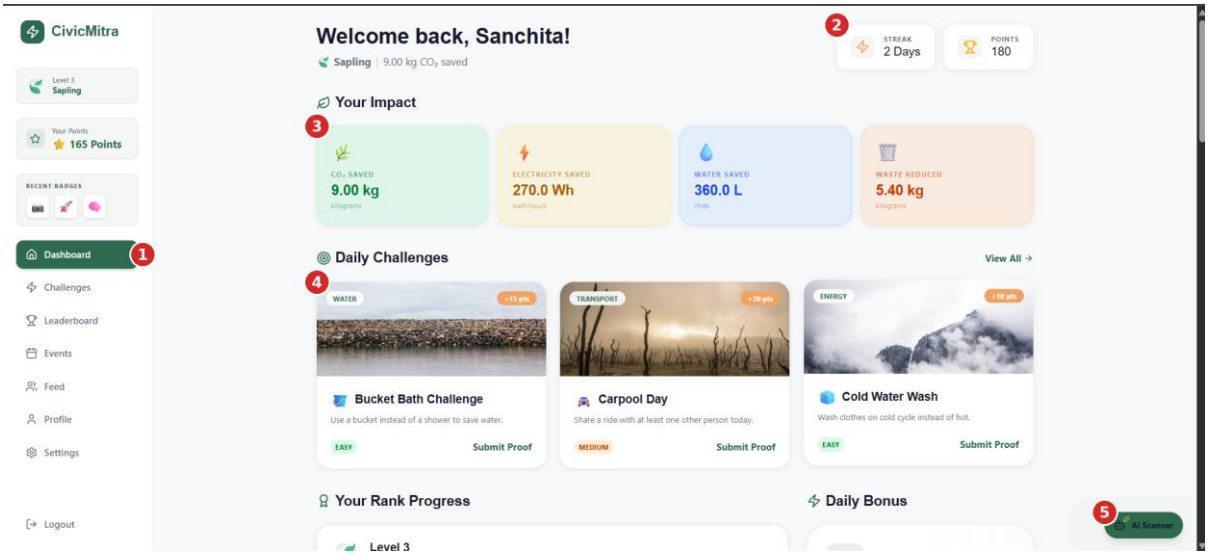
README - Setup and Run Instructions

Prerequisite: Node.js

- Install dependencies: npm install
- Set GEMINI_API_KEY in .env.local to your Gemini API key
- Run the app: npm run dev

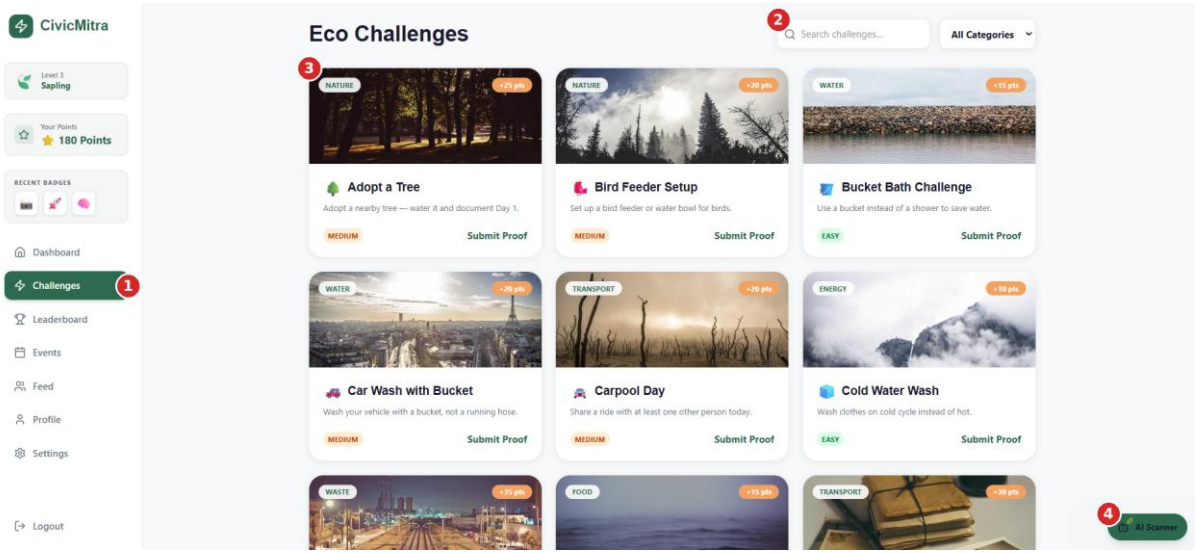
5. Prototype Screenshots

5.1 Dashboard



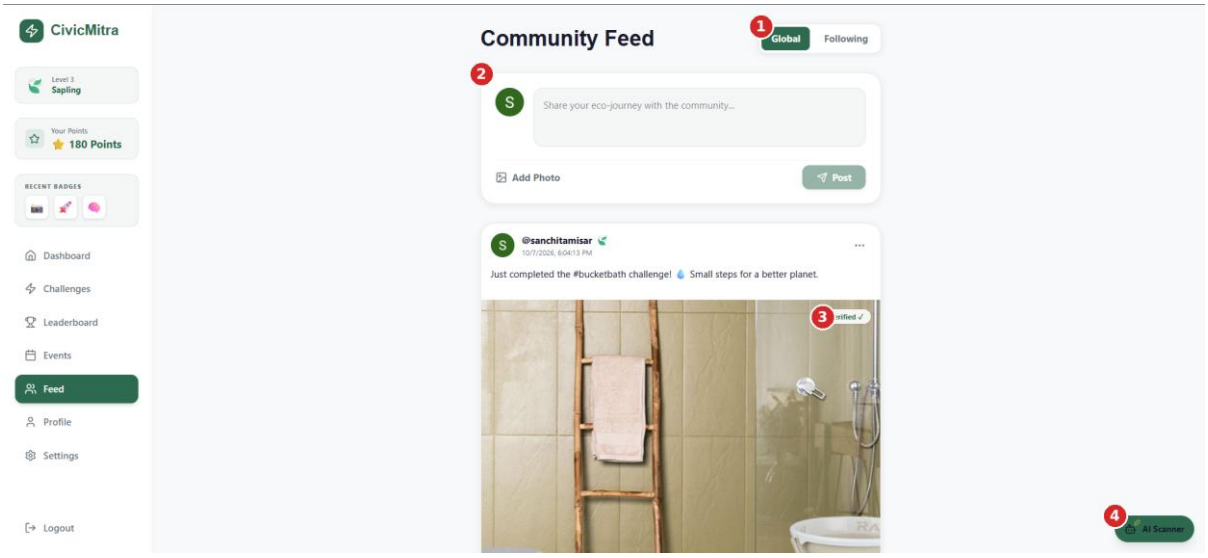
- 1 - Sidebar navigation
- 2 - Streak and points
- 3 - Your Impact cards (CO2, electricity, water, waste)
- 4 - Daily Challenges with Submit Proof
- 5 - AI Scanner button

5.2 Eco Challenges



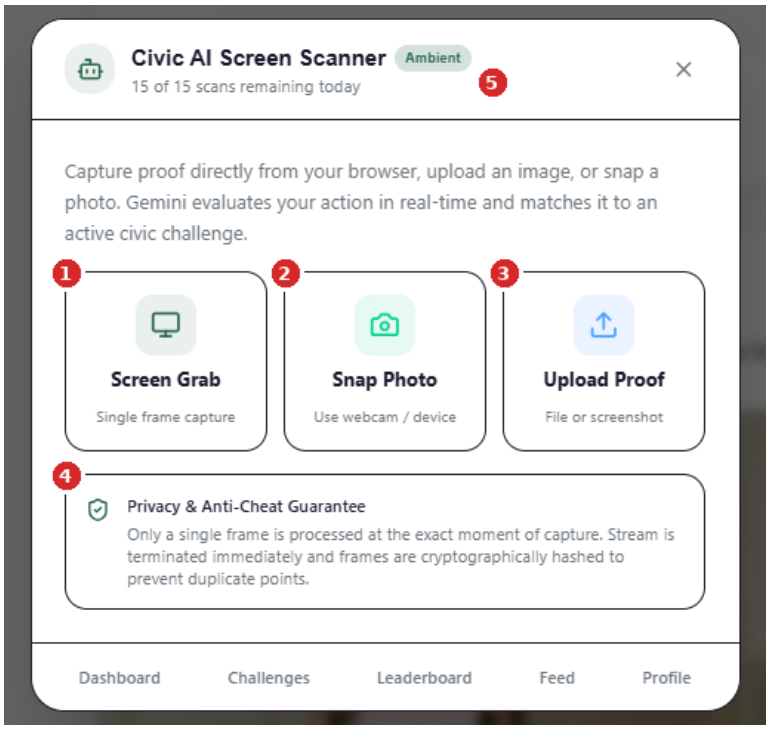
- 1 - Challenges menu (active)
- 2 - Search box and category filter
- 3 - Challenge cards with category, points and difficulty
- 4 - AI Scanner button

5.3 Community Feed



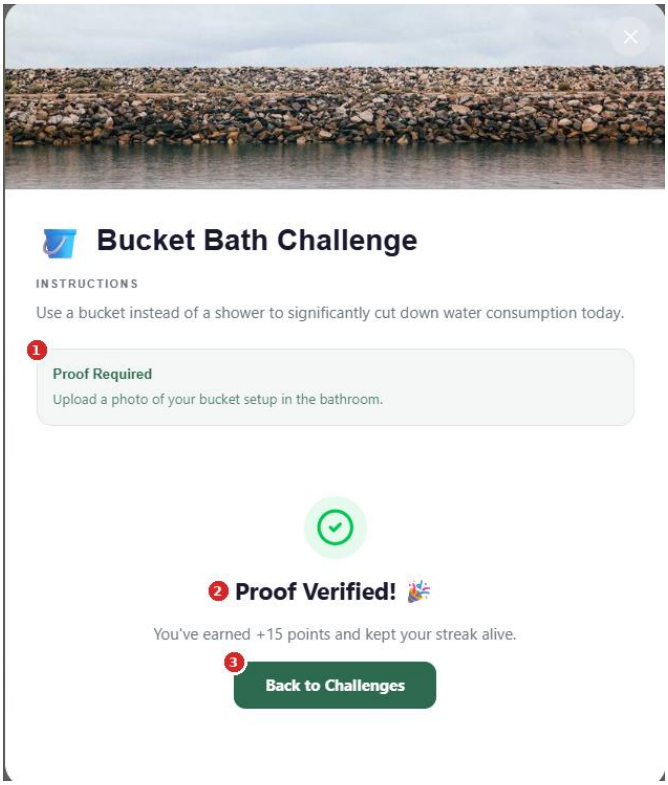
- 1 - Global / Following toggle
- 2 - Post box with Add Photo
- 3 - Verified badge on a completed challenge post
- 4 - AI Scanner button

5.4 Civic AI Screen Scanner



- 1 - Screen Grab
- 2 - Snap Photo
- 3 - Upload Proof
- 4 - Privacy & Anti-Cheat Guarantee
- 5 - Scans remaining today

5.5 Proof Verified



- 1 - Proof Required
- 2 - Proof Verified message with points earned
- 3 - Back to Challenges button